

Thermodynamics

Lecture 2A:

- Temperature & pressure
- Energy, Energy Transfer

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After this lecture

- Discuss definitions of density, temperature and pressure.
- Understand the concept of energy and its different forms.
- Learn about the nature of internal energy.
- Concept of energy transfer by heat and work.

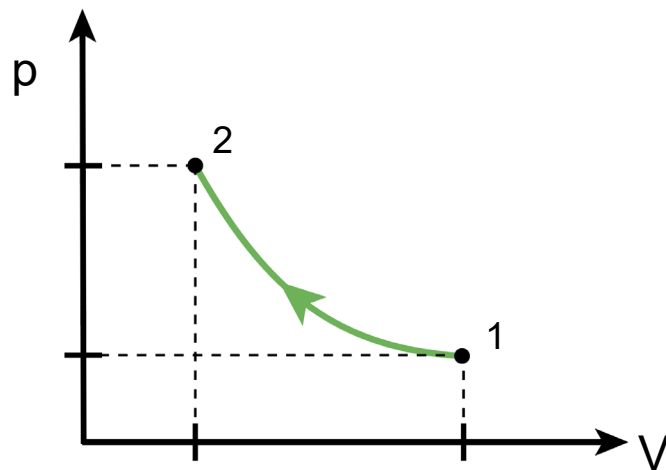
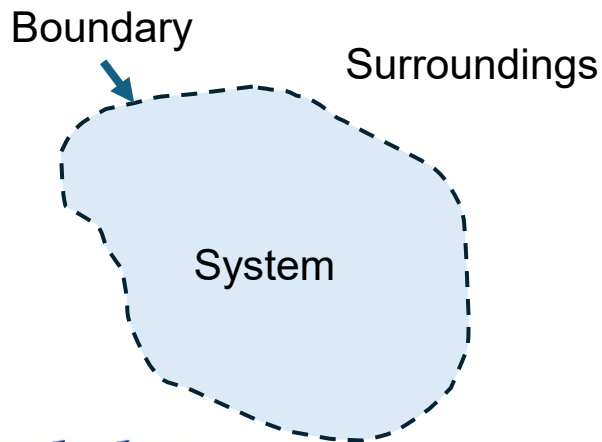
Learning objectives

What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. Comprehension:* Describe the first and second laws in the student's own language.
- III. Application:* Solve simple single-answer problems using the first law.
- IV. Analysis:* Solve problems requiring both the first and second laws.
- V. Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

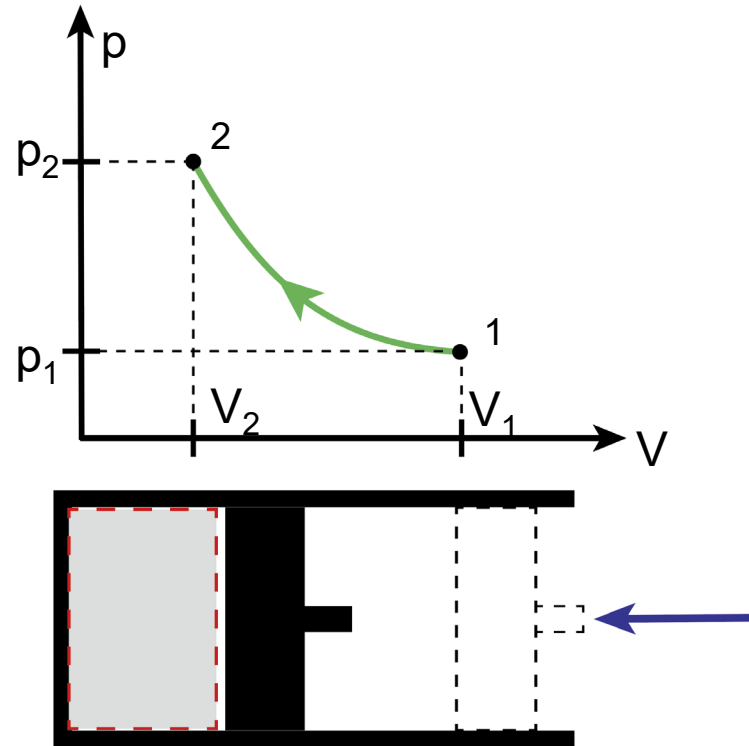
Recap Lecture 1B

- Define thermodynamic System: open, isolated, closed.
- Thermodynamic state and state properties.
- Thermodynamic processes

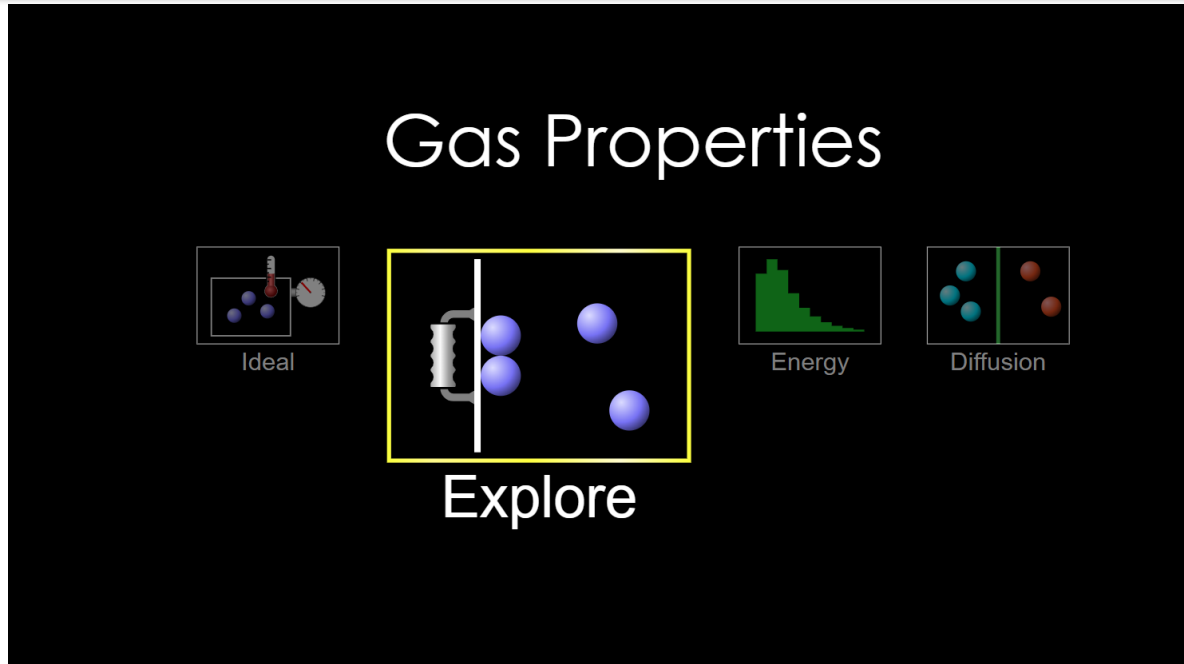


Thermodynamic process (II)

- Quasistatic process:
 - Process runs sufficiently slow so that systems passes through a series of equilibrium states.
- Non-quasistatic process:
 - Process runs too fast for equilibrium to be reached.

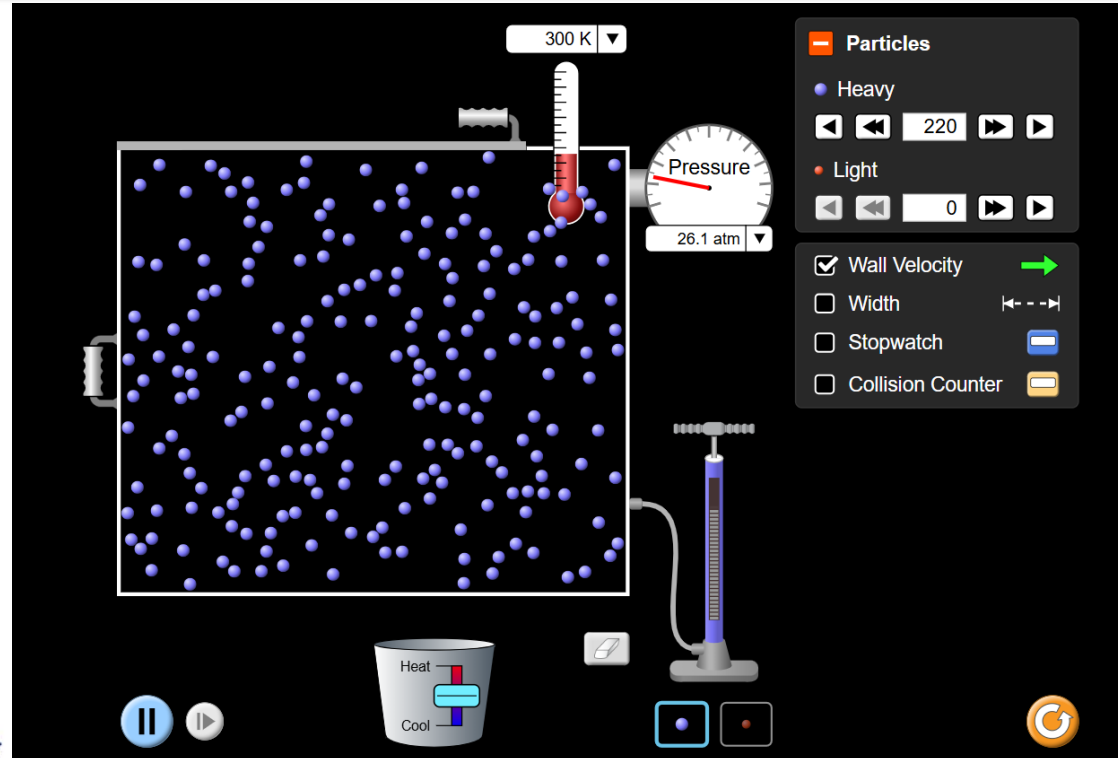


Simulation: Gas in a cylinder



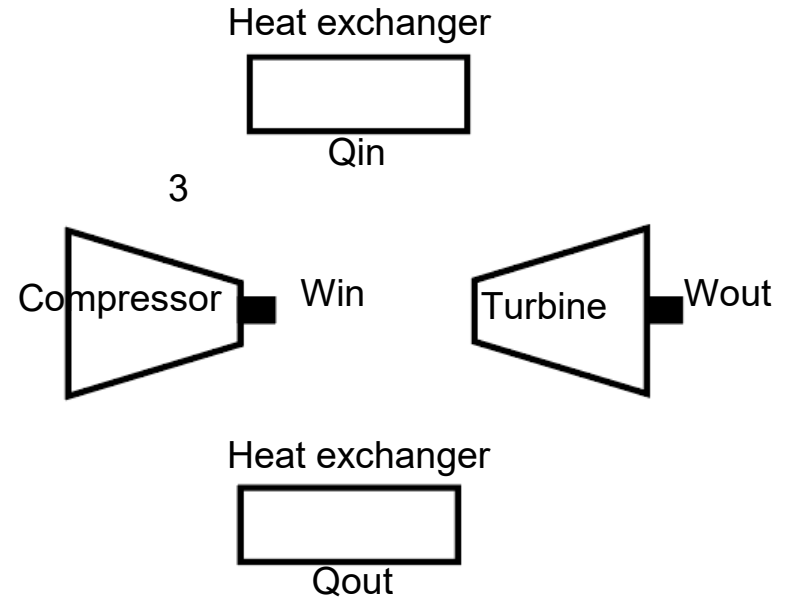
https://phet.colorado.edu/sims/html/gas-properties/latest/gas-properties_all.html

Simulation: Gas in a cylinder



Thermodynamic Cycles

- System undergoing a series of Th. processes and returning to the initial state.



Density

- Density: Mass per volume
- Specific Volume:
- Generally:
- Incompressible fluid:

Temperature

- A measure of how hot or cold
- **Celsius:** ice point 0°C , steam point 100°C
- **Fahrenheit:** ice point w salt 0°F , human body 96°F
- **Kelvin:** absolute zero 0 K

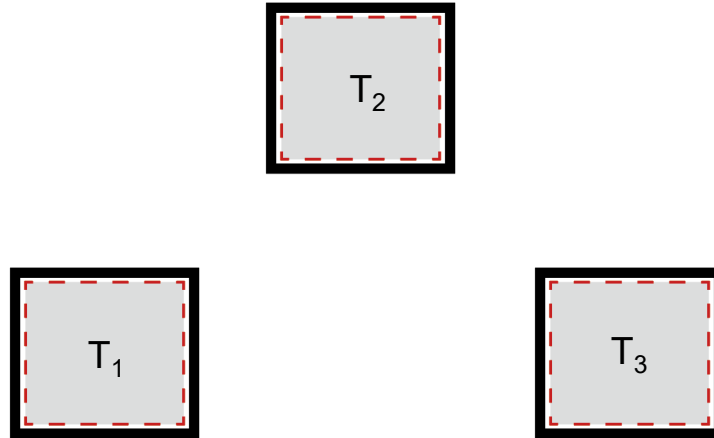
$$T(K) = T(^{\circ}\text{C}) + 273.15$$

$$T(K) = \frac{T(^{\circ}\text{F}) - 32}{1.8} + 273.15$$

$$\Delta T(K) = \Delta T(^{\circ}\text{C})$$

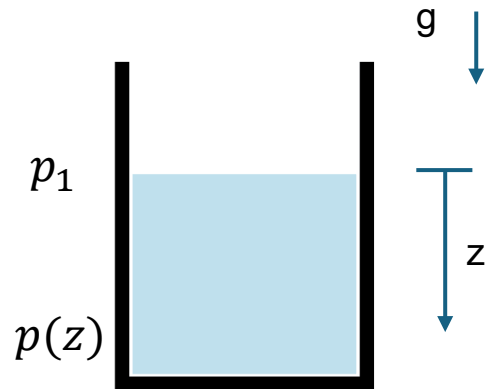
0th law of thermodynamics

- Thermal equilibrium: two objects at the same temperature.
- 0th law:

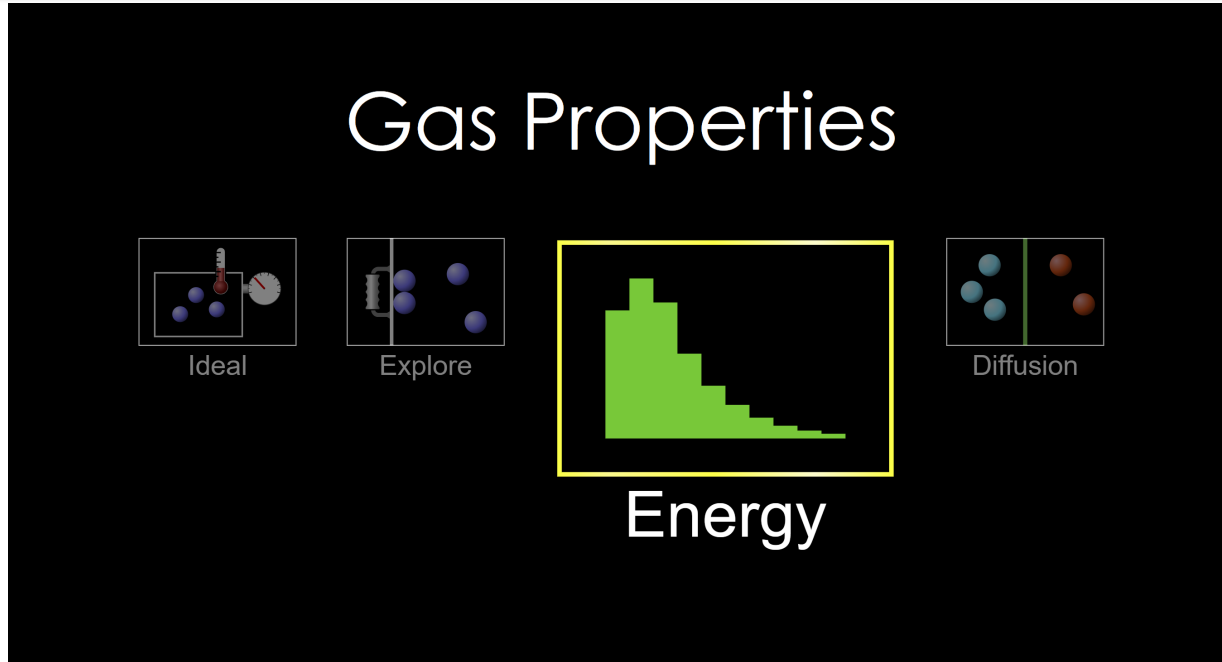


Pressure

- Normal force per unit area
- Absolute pressure p_{abs} relative to vacuum
- Gauge pressure:
- Vacuum pressure:
- External forces:

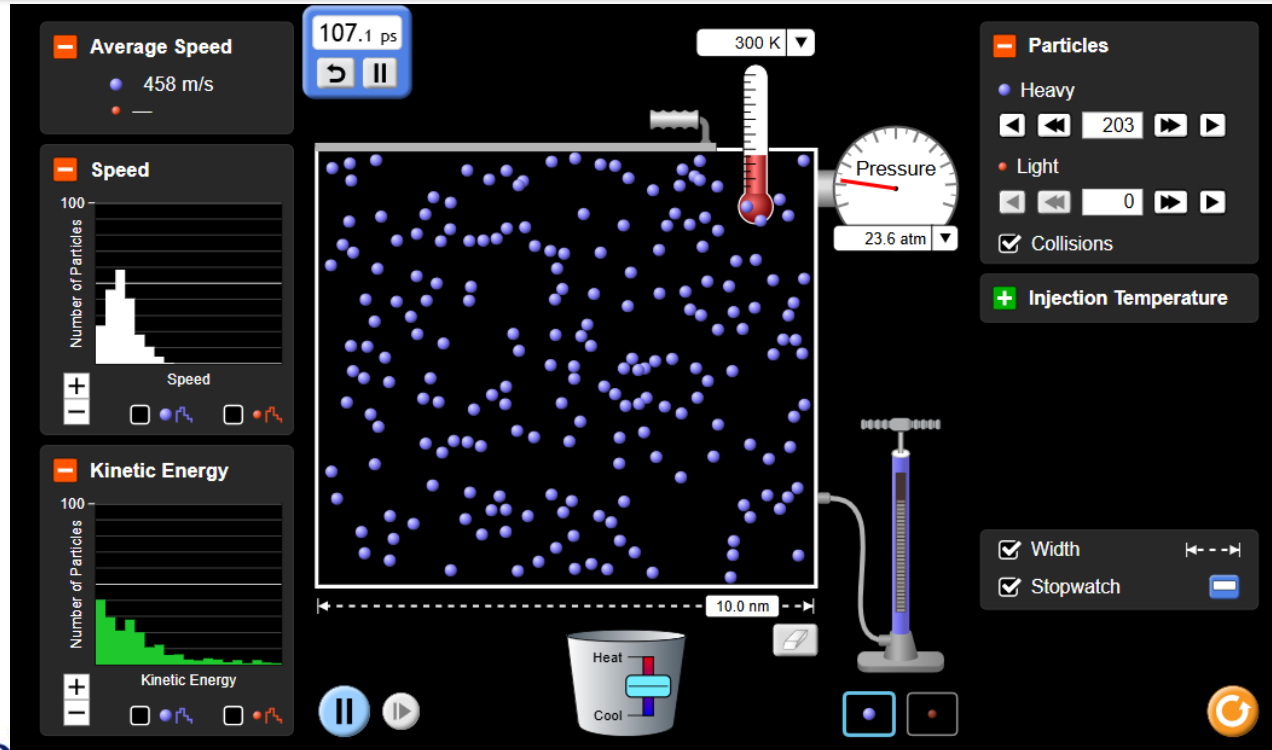


Simulation: Gas properties



https://phet.colorado.edu/sims/html/gas-properties/latest/gas-properties_all.html

Simulation: Gas properties



Energy

- An abstract mathematical idea. Energy is always conserved, cannot be generated, destroyed – rather converted from one form to another.
- No information on absolute energy rather changes in energy.
- Can be transferred from one system to another.

Forms of Energy

- Different forms of Energy:
 - Kinetic energy KE
 - Potential energy PE
 - Internal energy U
 - Electromagnetic energy
 - Binding energy in nucleus of atoms (weak, strong)

Kinetic energy

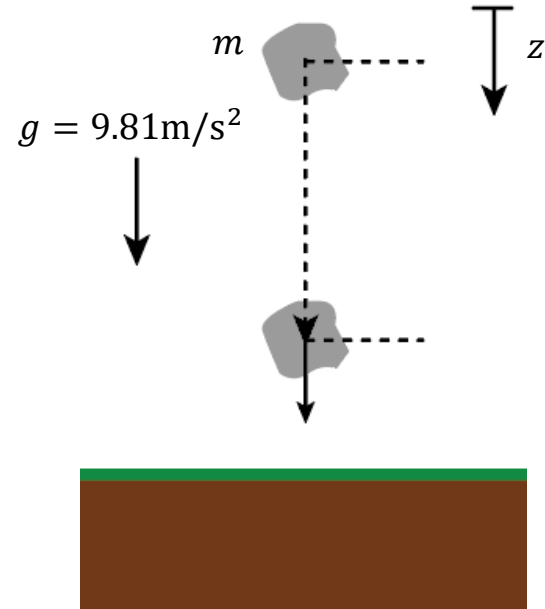
- Energy as a result of motion with respect to a reference frame.



Example of linear translation motion of a ball

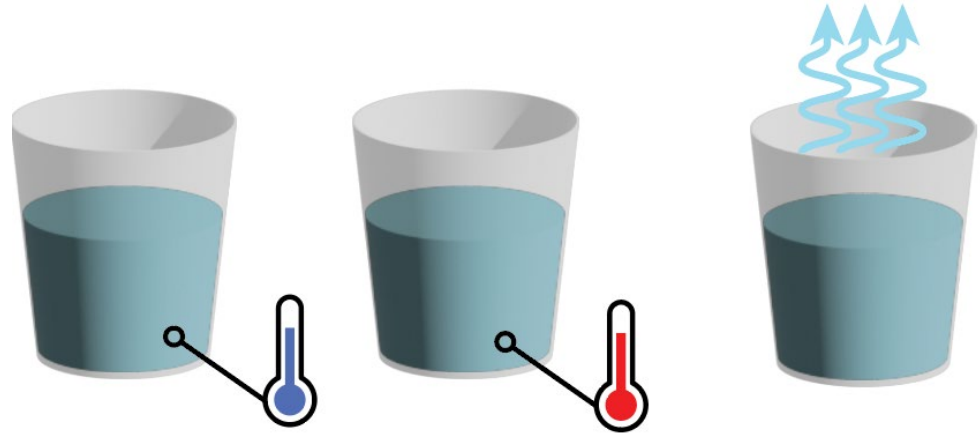
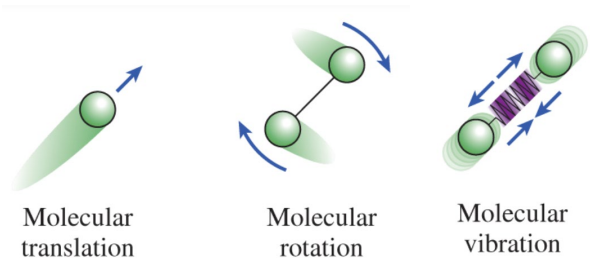
Potential energy

- Energy held by an object subject to a conservative force field (potential).
- Gravity:



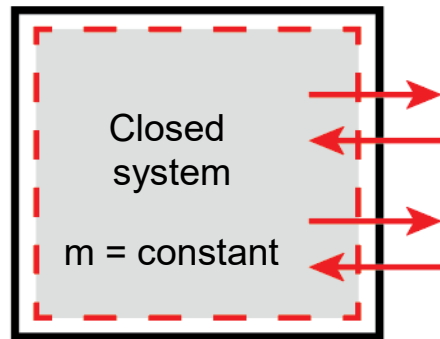
Internal Energy

- Microscopic nature of energy originating in the microscopic kinetic energy of matter (translational, rotational, vibrational), binding energy and chemical energy (potential).



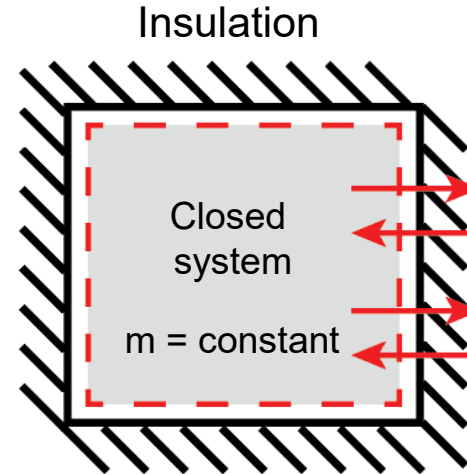
Energy transfer by heat

- Way of transferring energy to a closed system.
- Transfer of energy by virtue of a temperature difference. Direction of heat transfer from hot to cold.
- Terminology
 - Heat transfer
 - Heat transfer rate:
 - Heat transfer per mass:



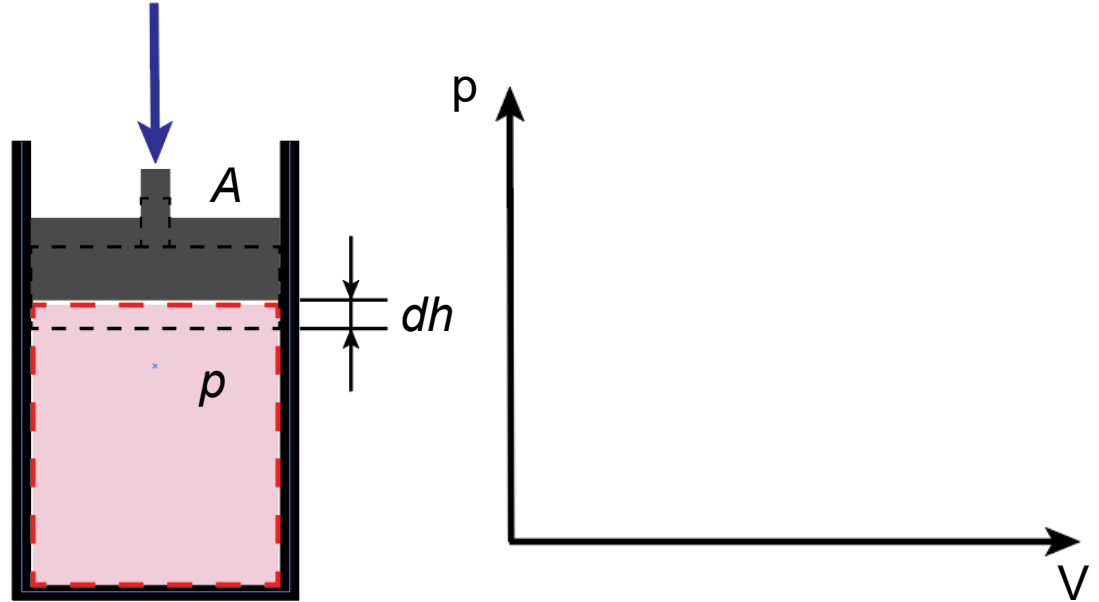
Adiabatic process

- A process where a system does not exchange heat with the surroundings.
- *Adiabatos*: not to be passed.



Energy transfer by work

- Work transfer on a closed system



Summary

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